

UniTS Anomaly Detection Report

ESA-Mission2 | Generated 2026-04-10 16:17

Dataset Summary

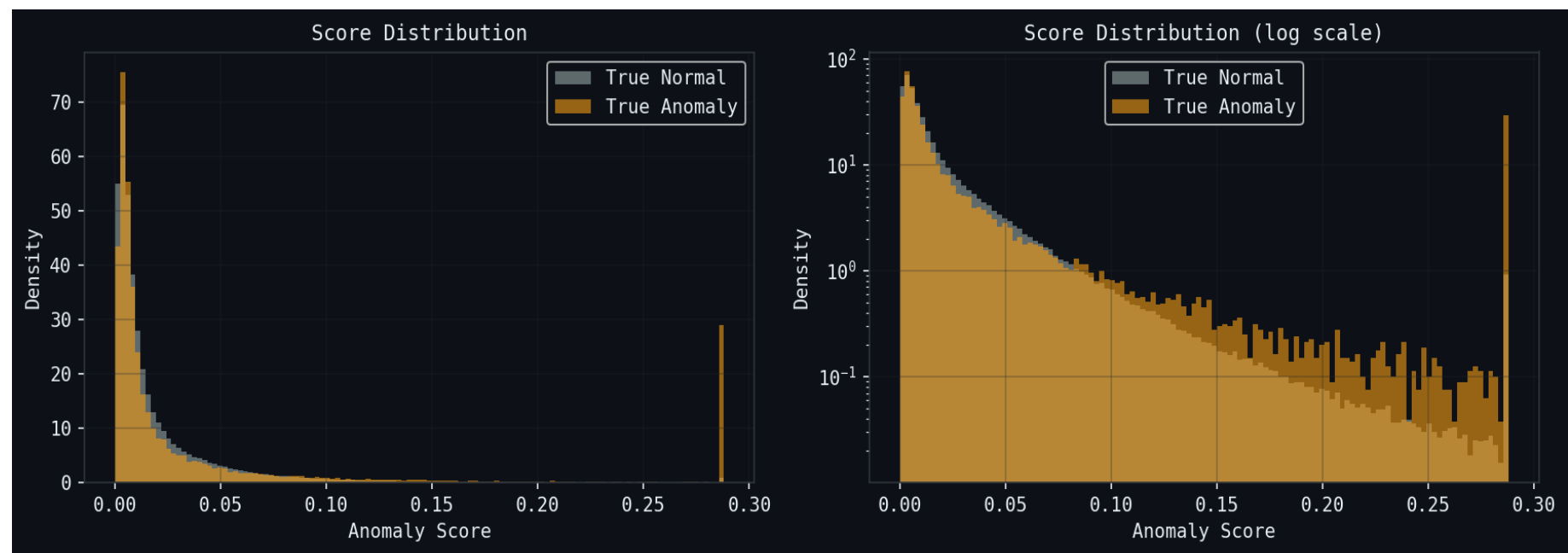
Total timesteps	786,240
Date range	2002-10-01 to 2003-06-30
Ground truth anomaly rate	4.19% (32,929 points)
Predicted anomaly rate	1.85% (14,583 points)

Detection Metrics

Metric	Value	Interpretation
Precision	0.9971	Of all predicted anomalies, 99.7% are real
Recall	0.4416	44.2% of all true anomaly points are detected
F1 Score	0.6121	Harmonic mean of precision and recall
Accuracy	0.9766	Overall correct classification rate
ROC-AUC	0.5356	Raw score discrimination (near-random — see note below)
Avg Precision	0.1202	Area under the Precision-Recall curve
True Positives	14,541	Anomaly points correctly flagged
False Positives	42	Normal points incorrectly flagged (0.0% of all normal points)
False Negatives	18,388	Anomaly points missed
True Negatives	753,269	Normal points correctly cleared

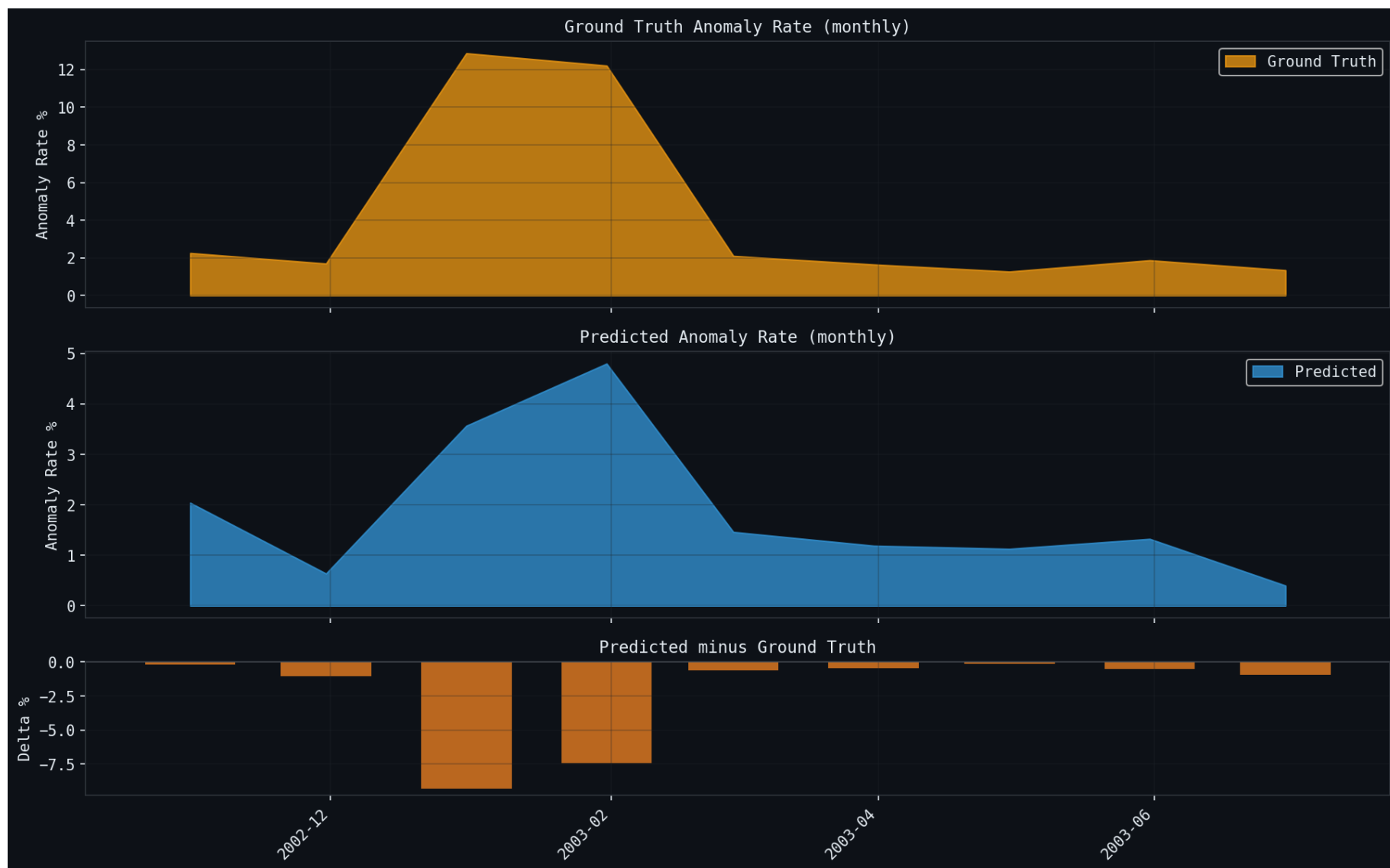
Note on F1 vs ROC-AUC: The F1 of 0.612 reflects binary detection quality under the UniTS point-adjust (PA) evaluation procedure, where detecting any point within an anomaly segment credits the full segment. The ROC-AUC of 0.536 reflects the raw anomaly_score used as a continuous ranking signal — it is near-random, meaning scores should not be used to prioritize or rank alerts. The binary flag is the actionable output.

Score Distribution



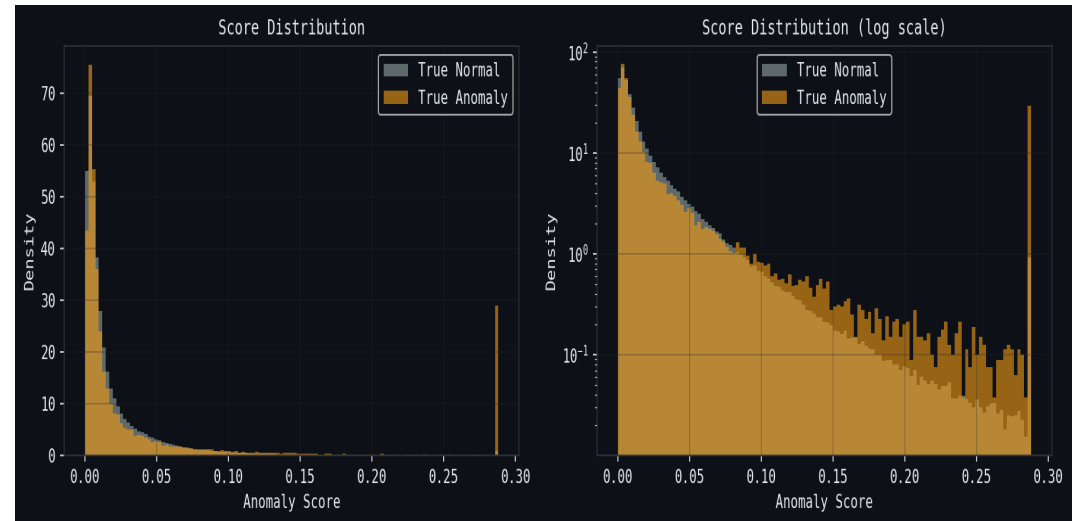
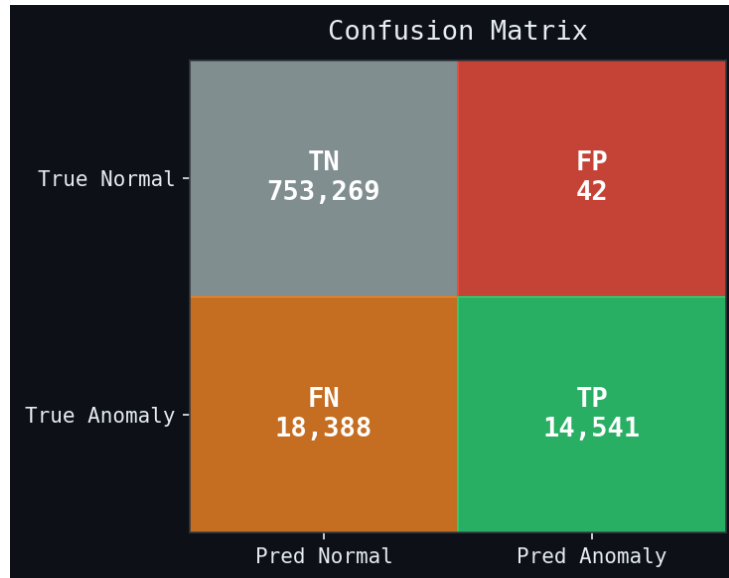
Score distributions clipped at the 99.5th percentile. Overlap between True Normal and True Anomaly distributions indicates the model relies on a fixed threshold rather than separating score distributions — consistent with ROC-AUC of 0.536.

Anomaly Timeline (monthly)

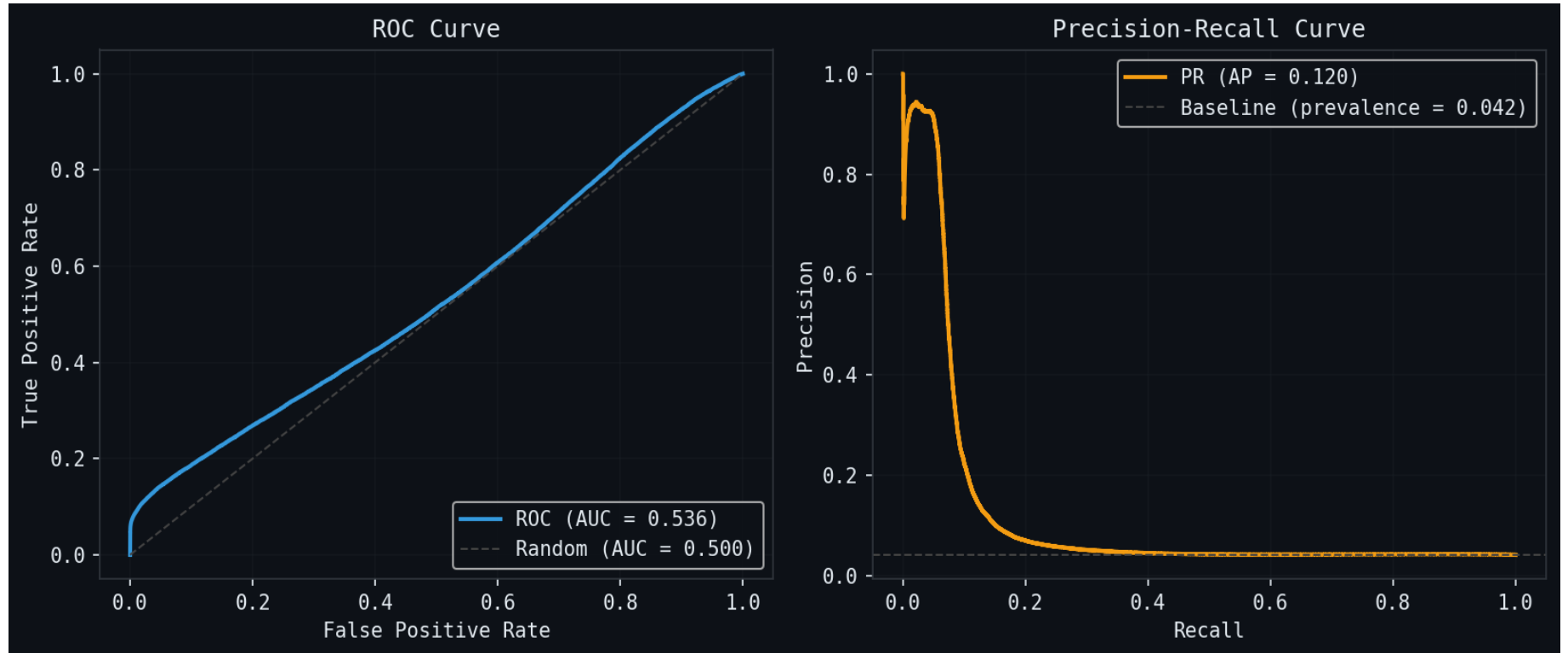


Most months show predicted rate slightly below ground truth, consistent with the 18,388 false negatives in the test period. Predicted and ground truth anomaly rates track closely through all major anomaly events.

Confusion Matrix & Score Distribution



ROC & Precision-Recall Curves



Both curves are computed on the raw `anomaly_score` (continuous). ROC-AUC = 0.536 (near-random). The binary `is_anomaly_predicted` flag (F1 = 0.612) is the operationally useful output.

Top False Positive Windows (pred=1, truth=0)

Total FP windows: 20 — showing top 20 by duration.

Start	End	Duration (pts)	Peak Score
2002-12-19 07:32:30	2002-12-19 07:33:30	3	59.7875
2003-01-02 22:26:30	2003-01-02 22:27:30	3	44.9232
2002-11-30 00:11:30	2002-11-30 00:12:30	3	36.2037
2002-12-18 00:56:30	2002-12-18 00:57:30	3	42.1494
2003-01-05 18:11:30	2003-01-05 18:12:30	3	38.8556
2003-01-26 18:11:30	2003-01-26 18:12:30	3	40.1622
2003-02-16 18:11:30	2003-02-16 18:12:30	3	37.6205
2003-02-02 10:46:30	2003-02-02 10:47:30	3	7.4670
2003-02-12 20:35:00	2003-02-12 20:35:30	2	29.1840
2002-10-27 04:50:30	2002-10-27 04:51:00	2	9.0299
2003-02-03 15:06:30	2003-02-03 15:07:00	2	20.1792
2002-10-21 00:50:30	2002-10-21 00:51:00	2	9.3538
2003-04-04 11:00:30	2003-04-04 11:01:00	2	23.2853
2003-02-02 10:49:30	2003-02-02 10:50:00	2	22.9734
2002-10-02 03:12:30	2002-10-02 03:12:30	1	4.0395
2002-10-05 09:28:30	2002-10-05 09:28:30	1	5.8339
2002-10-10 20:08:30	2002-10-10 20:08:30	1	3.8382
2002-11-03 18:32:30	2002-11-03 18:32:30	1	3.9606
2003-04-09 03:12:30	2003-04-09 03:12:30	1	5.9665
2003-05-21 03:12:30	2003-05-21 03:12:30	1	4.4345

Top False Negative Windows (pred=0, truth=1)

Total FN windows: 71 — showing top 20 by duration.

Start	End	Duration (pts)	Peak Score
2003-01-03 04:40:00	2003-01-04 12:34:00	3,829	0.2842
2002-12-31 05:06:30	2003-01-01 12:34:00	3,776	0.5647
2002-12-28 15:44:30	2002-12-29 16:52:30	3,017	0.3159
2002-12-16 13:26:30	2002-12-16 19:45:00	758	0.5059
2003-01-09 15:29:30	2003-01-09 18:38:30	379	0.1648
2003-05-23 01:28:00	2003-05-23 04:16:30	338	1.3042
2002-12-14 15:14:00	2002-12-14 17:53:30	320	1.6177
2003-06-12 16:01:00	2003-06-12 18:40:30	320	1.3952
2002-11-30 00:13:30	2002-11-30 02:50:30	315	1.5112
2002-12-10 15:11:00	2002-12-10 17:20:30	260	2.9440
2002-11-05 05:45:30	2002-11-05 07:37:00	224	1.3984
2002-12-02 02:20:30	2002-12-02 04:05:00	210	2.6506
2003-03-05 08:33:00	2003-03-05 10:13:30	202	0.2748
2002-12-18 01:00:30	2002-12-18 02:27:30	175	1.6061
2002-12-19 21:17:30	2002-12-19 22:39:30	165	0.3926
2002-11-15 17:45:30	2002-11-15 18:58:00	146	0.1581
2003-02-08 16:25:30	2003-02-08 17:35:30	141	0.1705
2002-12-19 04:44:30	2002-12-19 05:51:00	134	0.6796
2003-06-11 03:15:30	2003-06-11 04:19:00	128	2.8240
2002-10-03 06:33:00	2002-10-03 07:30:30	116	3.4542